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SHORT-PAPER

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Abstract

Recommender systems often rely on user visit trajectories, but the integration and representation of diverse side information remains a key challenge. Recent advances in large language models (LLMs) have enabled new strategies for enhancing this process. This study investigates how different types of side information support next Point-of-Interest (POI) recommendation, using a business-level dataset derived from Yelp. An LLM-based summarization pipeline is introduced to convert unstructured reviews and visual content into structured text via instruction-tuned models. These summaries, together with other business features, are each encoded into fixed-length embeddings. Based on these embeddings, four input configurations are constructed for BERT4Rec: trajectory-only, single feature categories, pairwise category combinations, and full combination. Our results show that side information consistently improves performance over the trajectory-only baseline, and their combinations exhibit useful synergies. These findings highlight the importance of modality-aware design and point toward adaptive fusion and selective use of side information. To support further research, we publicly release a multimodal POI recommendation dataset based on the Yelp Open Dataset.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

Point-of-Interest Recommendation, Side Information, Large Language Models, Multimodal Representation

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1 Introduction

With the rapid development of large language models (LLMs), increasing attention has been given to their use in recommender systems [12]. Prior studies show that enriching item representations with LLM-generated embeddings based on metadata, such as name and category, can improve recommendation outcomes [2]. However, it remains unclear whether incorporating additional business-side information, such as textual reviews or user-uploaded photos, can offer further improvements. This question is particularly relevant in domains like tourism and next point-of-interest (POI) recommendation, where rich contextual and semantic information may play a crucial role in user decision-making [21].

To address this, an existing sequential model is used as a starting point and initialized with enriched item representations. This paper focuses on improving item representations in next-POI recommendation by leveraging structured content generated via instruction-tuned LLMs. Specifically, user reviews and business photos are first summarized into structured text, then encoded together with other structured features using a unified embedding model. To better understand the impact of different side information, the following research questions (RQ) are posed:

- **RQ1:** How do different business-side information affect sequential POI recommendation performance in terms of accuracy and beyond-accuracy metrics?
- **RQ2:** When multiple business-related features are combined, do they lead to complementary gains?

To investigate these questions, an embedding-centric evaluation framework is developed to enable controlled comparisons of the impact of different business-side information in the context of next-POI recommendation. The contributions of this work are as follows:

- Construction and release of a Yelp-based POI dataset with diverse side information to support multimodal recommendation research. To our knowledge, it constitutes the first benchmark on the Yelp Open Dataset to include LLM-generated summaries of both reviews and photos[1].
- Introduction of a unified LLM-based pipeline that summarizes unstructured reviews and photos into concise, task-relevant textual representations. While many previous LLM-based studies focus on metadata, we construct four feature categories and analyze the synergy effects of embeddings generated through a shared encoder.

[1]The dataset is available at https://huggingface.co/datasets/wzehui/Yelp-Multimodal-Recommendation

- Development of an embedding-centric framework that decouples item representations from model architecture. This design enables controlled evaluation of business-side information across models and fusion strategies.

2 Related Work

Next-POI recommendation is a key application domain of recommender systems. It is commonly formulated as a sequential prediction task, where the goal is to infer the next location users are likely to visit based on their historical check-in sequences [11]. As a result, most POI recommendation studies continue to focus primarily on visit trajectory data [25].

To improve recommendation performance, existing works have incorporated business-side information, including geolocation [19, 26], textual reviews [13, 16], and visual content [4, 29]. Some studies have further explored integrating multiple types of business-side information to improve recommendation quality. For example, user-generated content (UGC) such as textual descriptions and images have been fused to enhance POI recommendation accuracy [15, 29]. However, limited attention has been given to systematically evaluating how different types of business-side information contribute to recommendation quality within a unified framework.

In the broader context of information-rich recommendation, numerous approaches have aimed to enhance *item representations* using business-side information [14]. Most recently, LLMs have emerged as a new means to integrate diverse side information into unified textual representations for downstream recommendation tasks [3, 23, 24]. Building on prior work that demonstrated the effectiveness of LLM-generated embeddings based on business metadata (e.g., name and category) for sequential recommendation [2], this study extends the investigation to richer business-side information. While earlier efforts focused on simple descriptors, the contribution of semantically complex information remains underexplored. To address this gap, unstructured information is converted into compact textual representations via LLMs and assessed jointly with structured information in a unified embedding-based framework.

3 Side Information Representation

To obtain different business-side information representations, an appropriate dataset is constructed and a unified embedding framework is adopted.

Data Preprocessing. Popular POI recommendation datasets such as Foursquare [30] and Gowalla [5] lack rich business-side information, which is crucial for analyzing their impact on sequential recommendation performance. To address this limitation, a POI recommendation dataset is constructed based on the Yelp Open Dataset[2], which includes extensive business-side information such as location, attributes, user reviews, and user-uploaded photos. A series of preprocessing steps is applied to user check-in data to ensure data quality and behavioral consistency[3]. These include:

- **Time Span Filtering:** Data from 2017 to 2019 is used to ensure sufficient check-in volume and to avoid disruptions in user mobility patterns caused by the COVID-19 pandemic.
- **Bot User Removal:** As Yelp logs are based on user-submitted reviews rather than verified check-ins, users exhibiting unrealistic travel itineraries are identified and removed.
- **Minimum Interaction Filtering:** A 10-core filtering strategy is applied by removing users and businesses with fewer than 10 check-ins, ensuring that behavioral sequences are long enough for meaningful modeling.

The statistics of the preprocessed POI dataset are summarized in Table 1.

Table 1: Statistics of the POI dataset with rich information

#Interactions	509,892	**Avg. User Seq. Length**	23.0
#Businesses	16,702	**Avg. Reviews per Business**	30.5
#Users	22,207	**Avg. Photos per Business**	9.0
Sparsity	99.86%	**Businesses with Photos**	80%

LLM-Based Summarization. Business reviews and photos are unstructured data, and embedding them directly may introduce noise or irrelevant signals [20]. Recent advances show that LLMs can summarize such information into meaningful representations for recommendation [12].

To extract salient and task-relevant information, a structured analytical summarization strategy via LLMs[4] is applied to both textual and visual content. For textual reviews, DeepSeek R1 [6] is used to aggregate all user reviews associated with a business into a structured summary. For visual content, GPT-4o [18] is employed to generate structured descriptions from all user-uploaded photos of a business.[5]

To enhance reliability, we adopt structured field-based summarization rather than free-text generation, which improves controllability and reduces hallucination risk. For visual inputs, we filter out low-quality or ambiguous content and only retain samples with clear semantic cues. The entire generation pipeline is fully reproducible, with all prompts and processing code publicly released to facilitate future research and improvements. The extracted fields are listed in Table 2, and the exact prompts are provided in the online material for reproducibility.

These structured summaries, along with other atomic features, are subsequently grouped by their information types to form feature categories for text embedding.

Unified Embedding Generation. Based on business-level atomic features, four feature categories are constructed for embedding generation:

- **Metadata:** Basic business descriptors such as name and category tags, which capture the general semantics of each POI [2].
- **Geolocation:** Precise location details (e.g., street address, city, coordinates) and multi-resolution[6] neighborhood representations based on H3 spatial indexing [8, 27].

[2]https://business.yelp.com/data/resources/open-dataset/

[3]Exact preprocessing details are provided in the accompanying online material alongside the released dataset.

[4]LLMs are selected based on their general performance as reported on the Chatbot Arena leaderboard (https://lmarena.ai).

[5]Visual summaries are generated only when captions and labels offer clear cues to minimize hallucination.

[6]Neighborhoods are encoded at multiple granularities (resolutions 5, 7, and 9).

Table 2: Structured content derived from reviews and photos.

Key	Description
`summary`	A brief overview of the overall review or visual environment.
`keywords`	Salient keywords extracted from textual reviews or visual content.
`sentiment score`	A polarity score from −1 (very negative) to 1 (very positive).
`sentiment confidence`	Estimated reliability of the sentiment score (range: 0–1).
`color tone`	Dominant indoor color palette observed in the photos.
`venue style`	Overall venue design including furniture, materials, and decor.
`food and drink style`	Visual characteristics of food and beverages.
`audience`	Visible customer demographics or dress patterns.
`special features`	Distinctive venue features (e.g., outdoor seating, live music).

- **User Feedback:** LLM-generated summaries of user reviews and rating statistics, providing a structured view of customer opinions.
- **Business Attributes:** Structured business-related attributes (e.g., ambience, pricing, accessibility) and LLM-generated visual content summaries, reflecting customers' visual impressions.

All feature categories are individually embedded using OpenAI's `text-embedding-ada-002`[7] to produce 1536-dimensional fixed-length representations. This design enables a representation-centric evaluation and helps isolate the contribution of each information type from potential fusion or modeling biases [28].

3.1 Embedding Configuration Strategies

Four embedding configurations are explored to evaluate the contribution of business-side information, as well as potential complementary effects.[8] These include:

- **Trajectory Only:** Item representations are learned solely from ID embeddings derived from user visit trajectories. This configuration serves as the baseline.
- **Single Categories:** Each embedding uses only one feature category: metadata, geolocation, user feedback, or business attributes. This facilitates the analysis of individual contributions.
- **Pairwise Category Combinations:** Embeddings from two feature categories are combined. Instead of enumerating all possible combinations, two representative pairs commonly adopted in existing research are selected. First, combining metadata and geolocation captures what a POI is and where it is located. This pairing reflects the core semantic and spatial identity of a business [10]. Second, combining user feedback and business attributes captures how the business is perceived by users both semantically and visually, as well as how it defines itself through structured attributes [17]. We limit our exploration to these two pairs, which represent the most complementary and interpretable combinations in existing literature.
- **Full Combination:** Combination of all available feature categories, evaluating the overall benefit of full information fusion.

According to these strategies, 8 embedding configurations are constructed. After dimensionality reduction[9], the embeddings serve as input to the selected sequential recommendation model.

4 Experimental Setup

To evaluate the impact of feature categories and their combinations, a modular setup is employed, in which the recommendation model remains fixed while the embedding input varies. The overall framework is illustrated in Figure 1. All experiments are conducted on NVIDIA A40 (48GB) and A800 (80GB)[10].

Recommendation Model. BERT4Rec [22] is adopted as the sequential recommendation model, as it demonstrated the best overall performance in prior work [2] and is widely used in next-item prediction tasks[11]. To incorporate semantically rich item representations, the ID-based embedding layer in BERT4Rec is replaced with pretrained textual embeddings derived from different types of business-side information. The pretrained embeddings are used to initialize the business embedding matrix. This design enables the model to leverage external semantic information while preserving the original architecture of BERT4Rec.

Data Splitting. The dataset is partitioned based on users' visitation histories in a temporal manner. Each user's full check-in sequence is treated as an indivisible unit: if all of a user's check-ins occur before a predefined cutoff date, the user is included in the training and validation sets; otherwise, the user's entire data is allocated to the test set. This design simulates a realistic cold-start scenario common in dynamic domains like tourism, where new users frequently enter the system. The final split results in an approximate 8:1:1 ratio of check-in records across the training, validation, and test sets, respectively.

Model Optimization. Hyperparameters for BERT4Rec are tuned separately for each input embedding configuration. The Optuna framework [1] is used to run 50 trials, optimizing NDCG@20 on the validation set. The hyperparameter search ranges and the corresponding optimal values are provided in the online material. For embedding configurations involving multiple feature categories, the embedding dimension is expanded to better preserve the information from each source. For the evaluation, the model is retrained on the union of training and validation sets. It is then evaluated on the test set 5 times, and the average performance is reported.

[7] https://platform.openai.com/docs/guides/embeddings/second-generation-models
[8] In this study, embeddings are combined via concatenation. Exploring more advanced fusion strategies is left for future work.
[9] We use Principal Component Analysis (PCA) in this work, following prior work [2] showing its effectiveness.
[10] The code is available at https://github.com/wzehui/LLM-Augmented-POIRec.git.
[11] To improve the generality of our results, we also experimented with other sequential models. Due to space limitations, we report only BERT4Rec results, where the synergy effects among modalities were most evident.

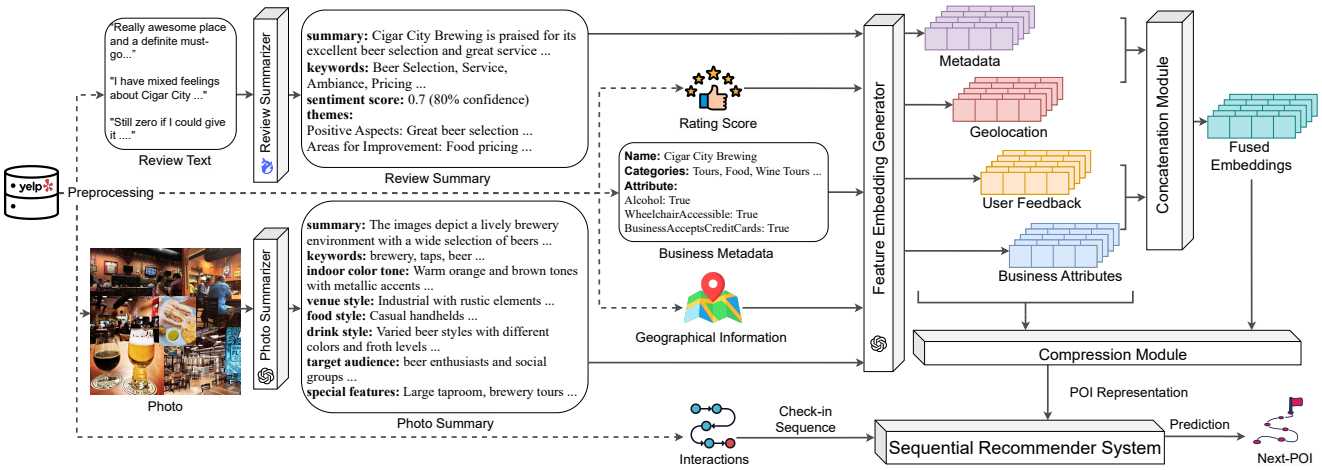

Figure 1: LLM-enhanced POI recommendation framework with multiple types of side information.

Evaluation Metrics. We evaluate recommendation performance using standard ranking accuracy metrics: NDCG, MRR, and Hit Rate. To more comprehensively assess the impact of different side information sources, we also report three beyond-accuracy metrics: Coverage [9], Serendipity [7], and Novelty [31]. We report results at a cutoff of 20 in the main paper, where differences between modality combinations are more pronounced. Results for other cutoffs are provided in the supplementary material.

5 Results and Analysis

To better understand the contribution of individual atomic features, the final composition of each feature category is determined by incrementally adding specific features and evaluating the downstream recommendation performance of the resulting embeddings. The performance of the optimal feature categories and their corresponding variants is presented in Table 3.

(1) **Metadata:** Compared to using category tags only (Metadata$^\diamond$), adding business names leads to consistent improvements across all accuracy metrics and a substantial increase in Coverage (+22.4%†)[12]. This suggests that business names may contain semantic or brand-specific cues that support niche relevance and personalization, helping to surface less popular, long-tail items. Both business name and category tags are retained in the final metadata category; (2) **Geolocation:** Based on precise location details, adding the neighborhood encoding (Geolocation$^\diamond$) substantially improves coverage (33.9%†), but leads to notable decreases in accuracy metrics, including NDCG (−11.0%†), and MRR (−13.9%†). While the neighborhood coding promote diversity by incorporating broader spatial context, it may also introduce spatial noise that impairs ranking precision. Thus, the neighborhood encoding is excluded from the final geolocation category; (3) **User Feedback:** Compared to using only the LLM-generated implicit textual descriptions (User Feedback$^\diamond$), adding explicit evaluation signals such as rating score and sentiment score[13] results in performance improvements of less than 10%. Despite the modest gain, the textual descriptions, rating scores, and sentiment scores are all retained in the final user feedback category due to their complementary semantic value and potential benefit to generalization; (4) **Business Attributes:** Relative to using visual content summaries alone (Business Attributes$^\diamond$), adding structured attributes significantly boosts coverage (+20.6%†) and yields slight improvements in accuracy metrics. This underscores the role of business-defined attributes in promoting item diversity. Both components are retained in the final business attributes configuration.

Using the best-performing feature category compositions identified above, single-category embeddings are generated respectively. These are further combined to form **Metadata+Geolocation**, **Feedback+Attributes**, and **Full Combination** embeddings. Together with the **Trajectory Only** baseline, the performance of these embeddings configurations is also presented in Table 3.

Overall, all four single-category embeddings outperform the **Trajectory Only** baseline across accuracy metrics and most beyond-accuracy metrics. These results answer **RQ1** by showing that incorporating business-side information improves both the accuracy and diversity of POI recommendations. While some atomic features contribute more than others, the findings suggest that enriching item representations beyond ID-based features can be beneficial.

Notably, **User Feedback** and **Business Attributes** achieve strong performance across all metrics, particularly in enhancing recommendation diversity. This demonstrates the advantage of integrating structured user opinions and visual summaries, which provide rich semantic cues and reflect subjective user experiences. In contrast, **Geolocation**, when used as a single feature category, performs less competitively. This may be attributed to the nature of the Yelp dataset, which is not based on actual trajectory data, resulting in weaker spatial correlations between user check-ins.

[12] † indicates a statistically significant difference ($p < 0.05$), based on a two-tailed independent samples t-test.

[13] Correlation analysis shows that the LLM-generated sentiment score and the rating score are correlated (Pearson correlation: 0.6776) and are therefore treated as a unified atomic feature.

Table 3: Impact of feature categories, their variants, and combinations on BERT4Rec performance (mean ± std). ***Rows are sorted in descending order based on NDCG@20.*** $^\diamond$ ***indicates a variant explored during the search for optimal atomic feature combinations.***

	NDCG@20	Hit Rate@20	MRR@20	Coverage@20	Serendipity@20	Novelty@20
Trajectory Only	0.0283 ± 0.0034	0.0679 ± 0.0063	0.0175 ± 0.0027	0.1897 ± 0.0391	0.0650 ± 0.0060	13.1788 ± 0.2813
Geolocation	0.0334 ± 0.0011	0.0824 ± 0.0025	0.0200 ± 0.0009	0.2090 ± 0.0487	0.0776 ± 0.0027	12.9907 ± 0.4142
Geolocation$^\diamond$	0.0297 ± 0.0030	0.0759 ± 0.0071	0.0172 ± 0.0022	0.2799 ± 0.0624	0.0724 ± 0.0056	13.1306 ± 0.3511
User Feedback	0.0338 ± 0.0019	0.0867 ± 0.0066	0.0195 ± 0.0010	0.2726 ± 0.0153	0.0833 ± 0.0064	13.1279 ± 0.1855
User Feedback$^\diamond$	0.0330 ± 0.0024	0.0811 ± 0.0050	0.0199 ± 0.0021	0.2732 ± 0.0388	0.0770 ± 0.0043	13.2962 ± 0.1818
Metadata	0.0342 ± 0.0043	0.0852 ± 0.0095	0.0204 ± 0.0036	**0.2994** ± 0.0338	0.0812 ± 0.0088	13.1447 ± 0.1985
Metadata$^\diamond$	0.0322 ± 0.0009	0.0815 ± 0.0059	0.0189 ± 0.0013	0.2445 ± 0.0423	0.0779 ± 0.0054	13.1616 ± 0.2662
Business Attributes	0.0351 ± 0.0026	0.0883 ± 0.0056	0.0206 ± 0.0018	0.2833 ± 0.0185	0.0850 ± 0.0058	**13.3698** ± 0.1401
Business Attributes$^\diamond$	0.0345 ± 0.0029	0.0843 ± 0.0046	0.0210 ± 0.0025	0.2350 ± 0.0433	0.0805 ± 0.0048	13.3123 ± 0.1939
Feedback+Attributes	0.0385 ± 0.0025	0.0927 ± 0.0037	0.0237 ± 0.0022	0.2750 ± 0.0361	0.0885 ± 0.0039	13.1929 ± 0.3353
Metadata+Geolocation	0.0386 ± 0.0015	0.0925 ± 0.0018	0.0238 ± 0.0013	0.2337 ± 0.0313	0.0882 ± 0.0016	13.2704 ± 0.1950
Full Combination	**0.0408** ± 0.0015	**0.1005** ± 0.0038	**0.0246** ± 0.0011	0.2935 ± 0.0201	**0.0966** ± 0.0036	13.2477 ± 0.2221

The answer to **RQ2** is reflected in the performance of feature category combinations. Both **Metadata+Geolocation** and **Feedback+Attributes**, outperform their respective single-category components. Using all available business-related features, the **Full Combination** configuration achieves the best overall performance across most metrics.[14] These results suggest that even without fusion-specific optimization, simple concatenation can activate useful synergies across feature categories. This richer representation space enables the recommendation of POIs that are both meaningful and less obvious (i.e., higher in novelty), addressing user intent from multiple perspectives.

6 Conclusion

This study investigated the impact of types of business-side information on next-POI recommendation using LLM-augmented embeddings within a multimodal recommendation setup. The experimental results confirm that incorporating metadata, geolocation, and LLM-summarized unstructured reviews and photos lead to improved performance compared to using visitation trajectories alone. Furthermore, combining certain business-related features produces synergies that further enhance recommendation quality.

This work opens up several opportunities for further exploration. Since the study is based on a single dataset in the tourism domain, extending the approach to other application contexts remains an important direction for validating generalizability. Furthermore, the current framework employs a static summarization strategy, which could be enhanced by incorporating user context or temporal dynamics during embedding generation to better capture recommendation intent. In terms of model evaluation, our study focuses on different combinations of side information within sequential recommendation frameworks, while reporting results based on BERT4Rec. We acknowledge that broader comparisons with other state-of-the-art models could provide a more comprehensive validation of the proposed embeddings. Lastly, exploring more advanced fusion strategies may further improve recommendation performance by better leveraging modality interactions. Taken together, these aspects highlight key areas for improvement, particularly in enhancing generalizability, context-awareness, and fusion robustness in multimodal recommendation.

[14] Even under equal embedding compression dimensions, full combination configuration outperforms all single-category inputs configurations in accuracy metrics, suggesting that the gains result from feature complementarity rather than increased representation size.

References

[1] Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. 2019. Optuna: A Next-generation Hyperparameter Optimization Framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* (Anchorage, AK, USA) *(KDD '19)*. 2623–2631. doi:10.1145/3292500.3330701

[2] Artun Boz, Wouter Zorgdrager, Zoe Kotti, Jesse Harte, Panos Louridas, Vassilios Karakoidas, Dietmar Jannach, and Marios Fragkoulis. 2025. Improving Sequential Recommendations with LLMs. *ACM Trans. Recomm. Syst.* (Jan. 2025). doi:10.1145/3711667

[3] Jizheng Chen, Kounianhua Du, Jianghao Lin, Bo Chen, Ruiming Tang, Weinan Zhang, and Yong Yu. 2024. ELCoRec: Enhance Language Understanding with Co-Propagation of Numerical and Categorical Features for Recommendation. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management* (Boise, ID, USA) *(CIKM '24)*. 259–269. doi:10.1145/3627673.3679789

[4] Ling Chen, Dandan Lyu, Shanshan Yu, and Gencai Chen. 2023. Multi-Level Visual Similarity Based Personalized Tourist Attraction Recommendation Using Geo-Tagged Photos. *ACM Trans. Knowl. Discov. Data* 17, 7, Article 92 (April 2023). doi:10.1145/3582015

[5] Eunjoon Cho, Seth A. Myers, and Jure Leskovec. 2011. Friendship and mobility: user movement in location-based social networks. In *Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (San Diego, California, USA) *(KDD '11)*. 1082–1090. doi:10.1145/2020408.2020579

[6] DeepSeek-AI. 2025. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. https://arxiv.org/abs/2501.12948

[7] Mouzhi Ge, Carla Delgado-Battenfeld, and Dietmar Jannach. 2010. Beyond accuracy: evaluating recommender systems by coverage and serendipity. In *Proceedings of the Fourth ACM Conference on Recommender Systems* (Barcelona, Spain) *(RecSys '10)*. 257–260. doi:10.1145/1864708.1864761

[8] Piotr Gramacki, Szymon Woźniak, and Piotr Szymański. 2021. gtfs2vec: Learning GTFS Embeddings for comparing Public Transport Offer in Microregions. In *Proceedings of the 1st ACM SIGSPATIAL International Workshop on Searching and Mining Large Collections of Geospatial Data* (Beijing, China) *(GeoSearch'21)*. 5–12. doi:10.1145/3486640.3491392

[9] Dietmar Jannach, Lukas Lerche, Iman Kamehkhosh, and Michael Jugovac. 2015. What recommenders recommend: an analysis of recommendation biases and possible countermeasures. *User Modeling and User-Adapted Interaction* 25, 5 (Dec. 2015), 427–491. doi:10.1007/s11257-015-9165-3

[10] Weiyu Ji, Xiangwu Meng, and Yujie Zhang. 2021. STARec: Adaptive Learning with Spatiotemporal and Activity Influence for POI Recommendation. *ACM Trans. Inf. Syst.* 40, 4, Article 65 (Nov. 2021). doi:10.1145/3485631

[11] Nicholas Lim, Bryan Hooi, See-Kiong Ng, Yong Liang Goh, Renrong Weng, and Rui Tan. 2022. Hierarchical Multi-Task Graph Recurrent Network for Next POI Recommendation. In *Proceedings of the 45th International ACM SIGIR Conference*

on Research and Development in Information Retrieval (Madrid, Spain) *(SIGIR '22)*. 1133–1143. doi:10.1145/3477495.3531989

[12] Jianghao Lin, Xinyi Dai, Yunjia Xi, Weiwen Liu, Bo Chen, Hao Zhang, Yong Liu, Chuhan Wu, Xiangyang Li, Chenxu Zhu, Huifeng Guo, Yong Yu, Ruiming Tang, and Weinan Zhang. 2025. How Can Recommender Systems Benefit from Large Language Models: A Survey. *ACM Trans. Inf. Syst.* 43, 2, Article 28 (Jan. 2025). doi:10.1145/3678004

[13] Zewei Lin and Liping Qu. 2023. Collaborative Knowledge-Aware Recommendation Based on Neighborhood Negative Sampling. *Information Systems* 115, Article 102207 (2023). doi:10.1016/j.is.2023.102207

[14] Qidong Liu, Jiaxi Hu, Yutian Xiao, Xiangyu Zhao, Jingtong Gao, Wanyu Wang, Qing Li, and Jiliang Tang. 2024. Multimodal Recommender Systems: A Survey. *ACM Comput. Surv.* 57, 2, Article 26 (Oct. 2024). doi:10.1145/3695461

[15] Xiaoqian Liu, Xiuyun Li, Yuan Cao, Fan Zhang, Xiongnan Jin, and Jinpeng Chen. 2023. Mandari: Multi-Modal Temporal Knowledge Graph-aware Sub-graph Embedding for Next-POI Recommendation. In *2023 IEEE International Conference on Multimedia and Expo (ICME)*. 1529–1534. doi:10.1109/ICME55011.2023.00264

[16] Pramit Mazumdar, Bidyut Kr. Patra, and Korra Sathya Babu. 2020. Cold-start Point-of-interest Recommendation through Crowdsourcing. *ACM Trans. Web* 14, 4, Article 19 (Aug. 2020). doi:10.1145/3407182

[17] Lyndon Nixon. 2022. Do DMOs Promote the Right Aspects of the Destination? A Study of Instagram Photography with a Visual Classifier. In *Information and Communication Technologies in Tourism 2022*, Jason L. Stienmetz, Berta Ferrer-Rosell, and David Massimo (Eds.). 174–186.

[18] OpenAI. 2024. GPT-4o System Card. https://arxiv.org/abs/2410.21276

[19] Yifang Qin, Yifan Wang, Fang Sun, Wei Ju, Xuyang Hou, Zhe Wang, Jia Cheng, Jun Lei, and Ming Zhang. 2023. DisenPOI: Disentangling Sequential and Geographical Influence for Point-of-Interest Recommendation. In *Proceedings of the Sixteenth ACM International Conference on Web Search and Data Mining* (Singapore, Singapore) *(WSDM '23)*. 508–516. doi:10.1145/3539597.3570408

[20] Ruihong Qiu, Sen Wang, Zhi Chen, Hongzhi Yin, and Zi Huang. 2021. CausalRec: Causal Inference for Visual Debiasing in Visually-Aware Recommendation. In *Proceedings of the 29th ACM International Conference on Multimedia* (Virtual Event, China) *(MM '21)*. 3844–3852. doi:10.1145/3474085.3475266

[21] Pablo Sánchez and Alejandro Bellogín. 2022. Point-of-Interest Recommender Systems Based on Location-Based Social Networks: A Survey from an Experimental Perspective. *ACM Comput. Surv.* 54, 11s, Article 223 (Sept. 2022). doi:10.1145/3510409

[22] Fei Sun, Jun Liu, Jian Wu, Changhua Pei, Xiao Lin, Wenwu Ou, and Peng Jiang. 2019. BERT4Rec: Sequential Recommendation with Bidirectional Encoder Representations from Transformer. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management* (Beijing, China) *(CIKM '19)*. 1441–1450. doi:10.1145/3357384.3357895

[23] Jiahao Tian, Zhenkai Wang, Jinman Zhao, and Zhicheng Ding. 2024. MMREC: LLM Based Multi-Modal Recommender System. In *2024 19th International Workshop on Semantic and Social Media Adaptation & Personalization (SMAP)*. 105–110. doi:10.1109/SMAP63474.2024.00028

[24] Shirui Wang, Bohan Xie, Ling Ding, Xiaoying Gao, Jianting Chen, and Yang Xiang. 2024. SeCor: Aligning Semantic and Collaborative Representations by Large Language Models for Next-Point-of-Interest Recommendations. In *Proceedings of the 18th ACM Conference on Recommender Systems* (Bari, Italy) *(RecSys '24)*. 1–11. doi:10.1145/3640457.3688124

[25] Zehui Wang, Wolfram Höpken, and Dietmar Jannach. 2025. A survey on point-of-interest recommendations leveraging heterogeneous data. *Information Technology & Tourism* 27, 1 (2025), 29–73. doi:10.1007/s40558-024-00301-3

[26] Zhaobo Wang, Yanmin Zhu, Chunyang Wang, Wenze Ma, Bo Li, and Jiadi Yu. 2023. Adaptive Graph Representation Learning for Next POI Recommendation. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Taipei, Taiwan) *(SIGIR '23)*. 393–402. doi:10.1145/3539618.3591634

[27] Szymon Woźniak and Piotr Szymański. 2021. hex2vec: Context-Aware Embedding H3 Hexagons with OpenStreetMap Tags. In *Proceedings of the 4th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery* (Beijing, China) *(GEOAI '21)*. 61–71. doi:10.1145/3486635.3491076

[28] Yueqi Xie, Peilin Zhou, and Sunghun Kim. 2022. Decoupled Side Information Fusion for Sequential Recommendation. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Madrid, Spain) *(SIGIR '22)*. 1611–1621. doi:10.1145/3477495.3531963

[29] Yang Xu, Gao Cong, Lei Zhu, and Lizhen Cui. 2024. MMPOI: A Multi-Modal Content-Aware Framework for POI Recommendations. In *Proceedings of the ACM Web Conference 2024* (Singapore, Singapore) *(WWW '24)*. 3454–3463. doi:10.1145/3589334.3645449

[30] Dingqi Yang, Daqing Zhang, Zhiyong Yu, and Zhiwen Yu. 2013. Fine-grained preference-aware location search leveraging crowdsourced digital footprints from LBSNs. In *Proceedings of the 2013 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Zurich, Switzerland) *(UbiComp '13)*. 479–488. doi:10.1145/2493432.2493464

[31] Tao Zhou, Zoltán Kuscsik, Jian-Guo Liu, Matúš Medo, Joseph Rushton Wakeling, and Yi-Cheng Zhang. 2010. Solving the apparent diversity-accuracy dilemma of recommender systems. *Proceedings of the National Academy of Sciences* 107, 10 (2010), 4511–4515. doi:10.1073/pnas.1000488107